



DIGESTION & ABSORPTION OF NUTRIENTS IN THE GASTROINTESTINAL TRACT

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Lecturer

DIKIOHS, DUHS.

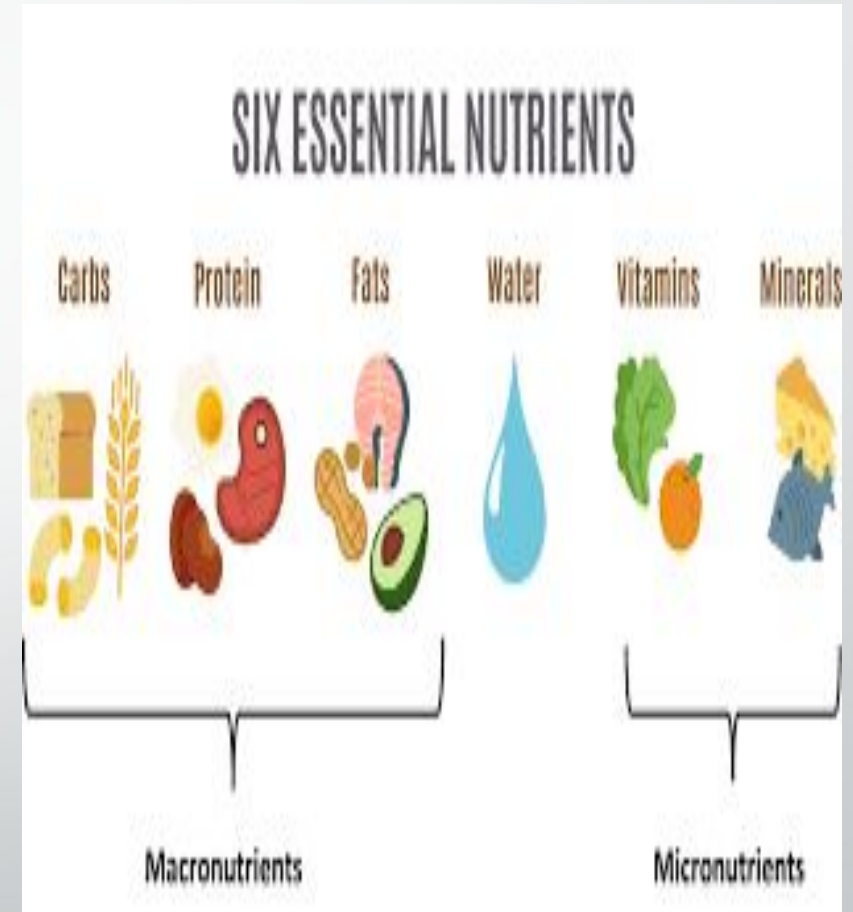
LEARNING OBJECTIVES

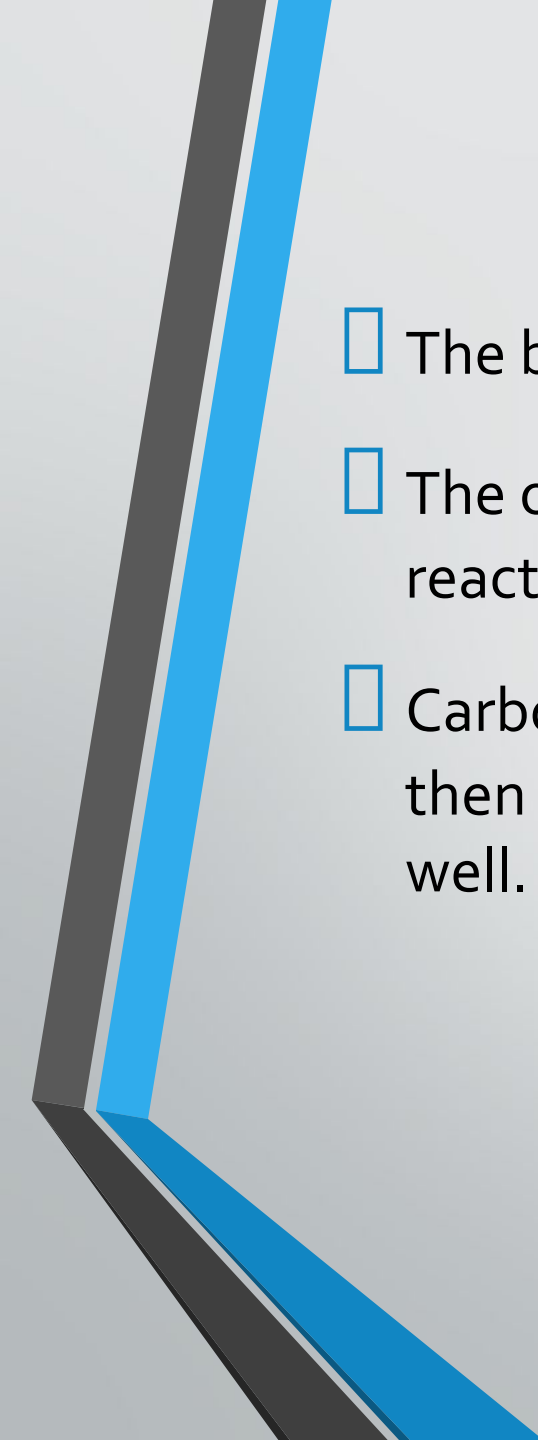
By the end of this lecture, students should be able to:

- ✓ Identify the basic components of the diet.
- ✓ Describe the overall process of digestion & absorption.
- ✓ Identify the major anatomical structures of the gastrointestinal tract involved in digestion & absorption including the mouth, stomach, small intestine, pancreas, liver & the large intestine.
- ✓ Describe the mechanical & chemical processes of digestion that occur in different regions of the GI tract.
- ✓ Identify the final end-products formed as a result of digestion of carbohydrates, proteins & fats.
- ✓ Identify the transport mechanisms involved in the uptake of these basic nutrients(carbohydrates, proteins, fats).
- ✓ Describe the digestion, emulsification, absorption & transport of lipids including the role of bile salts, bile micelles & chylomicrons.

INTRODUCTION

- The major foods on which the body lives are carbohydrates, fats & proteins.
- In addition to these vital nutrients, small quantities of substances like vitamins & minerals are also essential.
- Carbohydrates, fats & proteins cannot be absorbed in their natural forms through the gastrointestinal mucosa.
- The preliminary digestion of these nutrients is very important to make them beneficial for the body.



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- The basic digestion process for all the three types of food is ***hydrolysis***.
 - The only difference is the types of enzymes required for the hydrolysis reaction for each type of food.
 - Carbohydrates, fats & proteins are first digested into smaller compounds, then they are absorbed along with water, electrolytes & other substances as well.

DIGESTION OF FOOD BY HYDROLYSIS

HYDROLYSIS OF CARBOHYDRATES

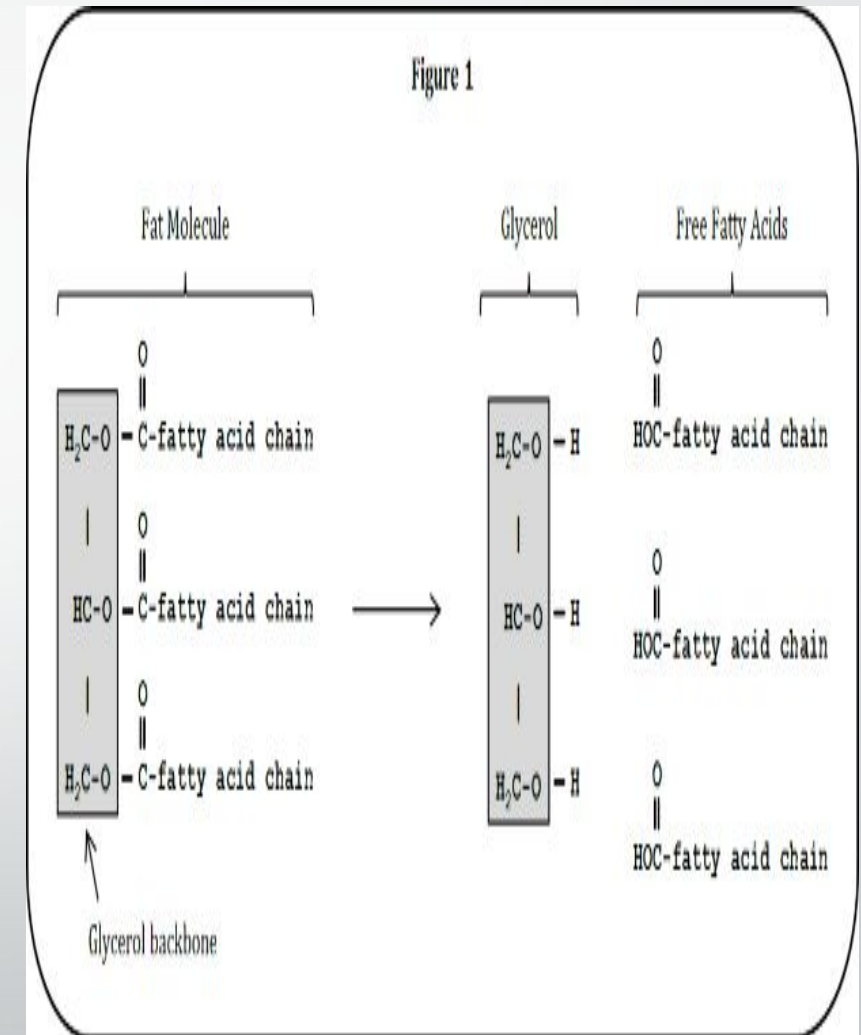
- Almost all the carbohydrates content of the diet (either polysaccharides or disaccharides) are bound to each other by condensation.
- Hydrolysis is a process in which a hydrogen ion (H^+) has been removed from one monosaccharide & a hydroxyl ion (OH^-) has been removed from the other.
- The two monosaccharides combine with each other at these sites of removal.
- The H^+ & the OH^- ions are then combined to form water.



$R'' - R'$ is a disaccharide.

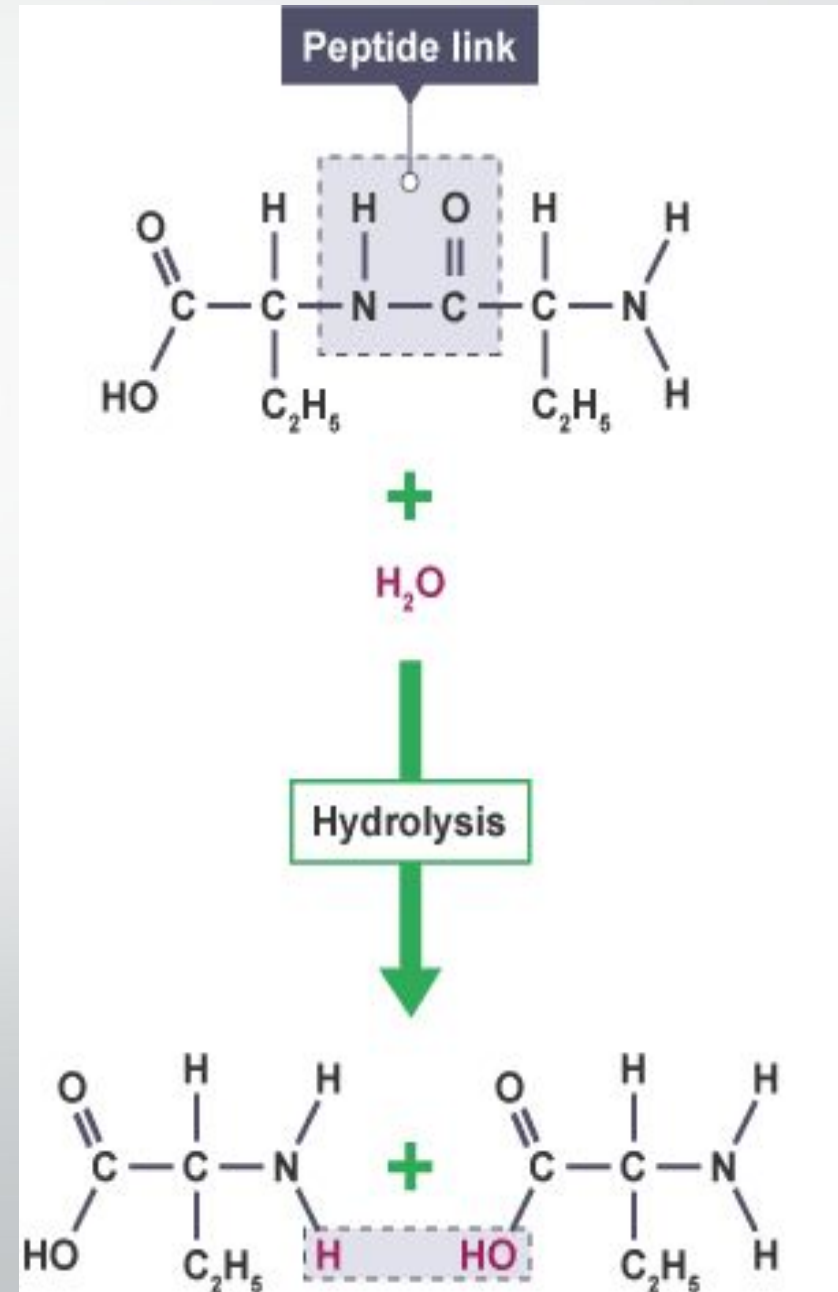
HYDROLYSIS OF FATS

- Almost the entire fat portion of the diet consists of triglycerides which are the combinations of three fatty acid molecules condensed with a single glycerol molecule.
- During condensation, three molecules of water are formed.
- Hydrolysis of fats is a reverse process in which fat-digesting enzymes return three molecules of water to the triglyceride molecule & split the fatty acids molecules away from glycerol.



HYDROLYSIS OF PROTEINS

- Proteins are formed from multiple amino-acids bound together by **peptide linkages**.
- At each peptide linkage, one OH^- has been removed from one amino-acid & one H^+ has been removed from the other.
- In this way, the successive amino-acids in the protein chain are also bound together by **condensation**.
- The digestion occurs by the reverse process i.e. **Hydrolysis**.
- In hydrolysis, the proteolytic enzymes return H^+ & OH^- from water molecules to the protein molecules to split them into constituent amino-acids.



CARBOHYDRATES IN DIET

- Human diet contains three types of carbohydrates:
 - i. Polysaccharides
 - ii. Disaccharides
 - iii. Monosaccharides

POLYSACCHARIDES

- Polysaccharides in our diet are ***glycogen, amylose & amylopectin.***
- These polysaccharides are present in the form of ***starch*** which is a glucose polymer.
- Glycogen is present in non-vegetarian diet whereas amylose & amylopectin are available in vegetarian diet(plant origin).

DISACCHARIDES

□ Disaccharides available in the diet are:

i. Sucrose

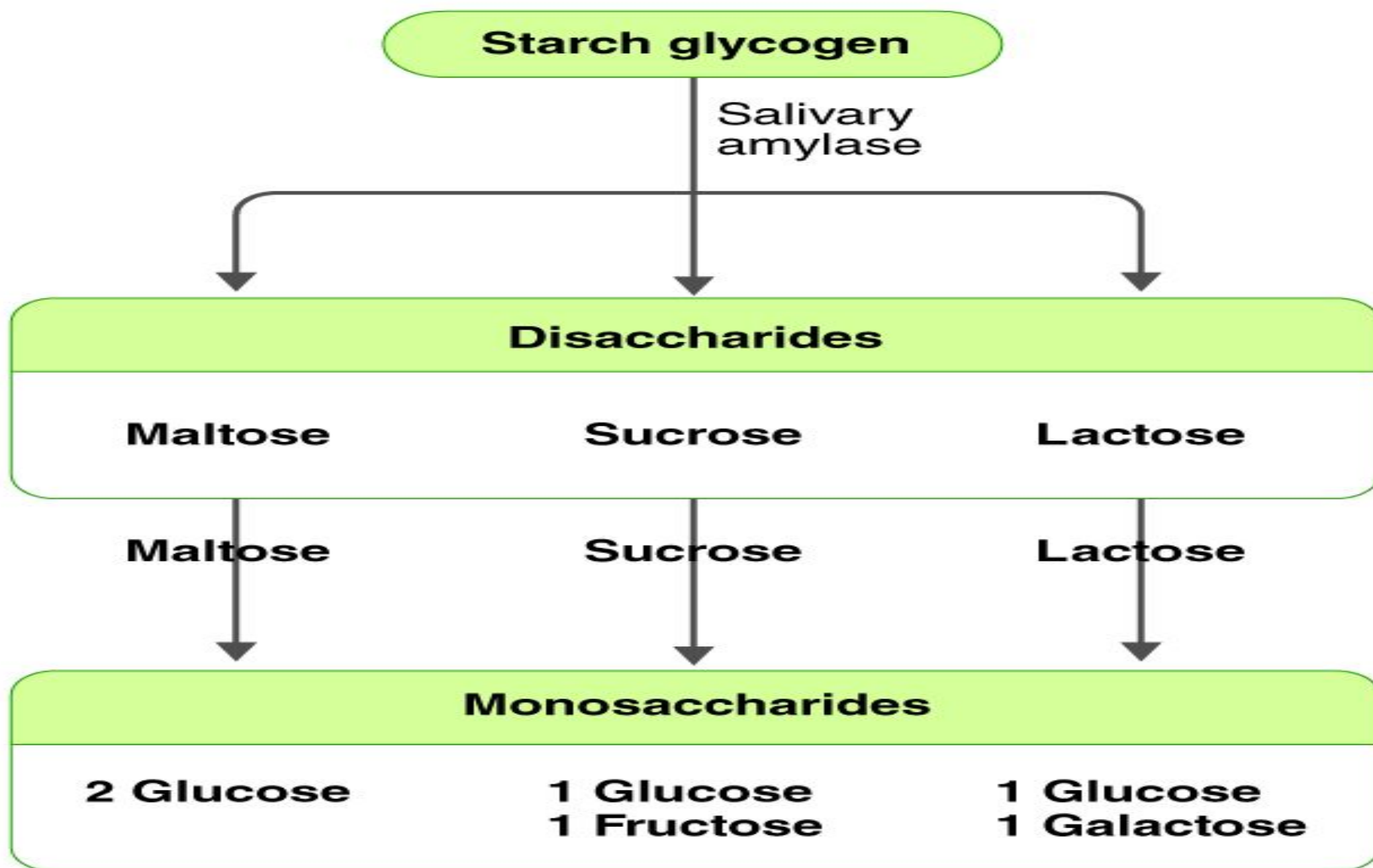
ii. lactose.

□ Sucrose is the combination of glucose & fructose molecules & is known as ***table sugar*** or ***cane sugar***.

□ Lactose is formed by the combination of glucose molecules with galactose molecules & is present in the ***milk***.

- Monosaccharides consumed in human diet are mostly ***glucose & fructose***.
- Other carbohydrates present in human diet are lactic acid, pyruvic acid, pectins, dextrans etc.
- Human diet also contains large amounts of ***cellulose*** which cannot be digested by human GI tract so it is not considered as a food for human beings.

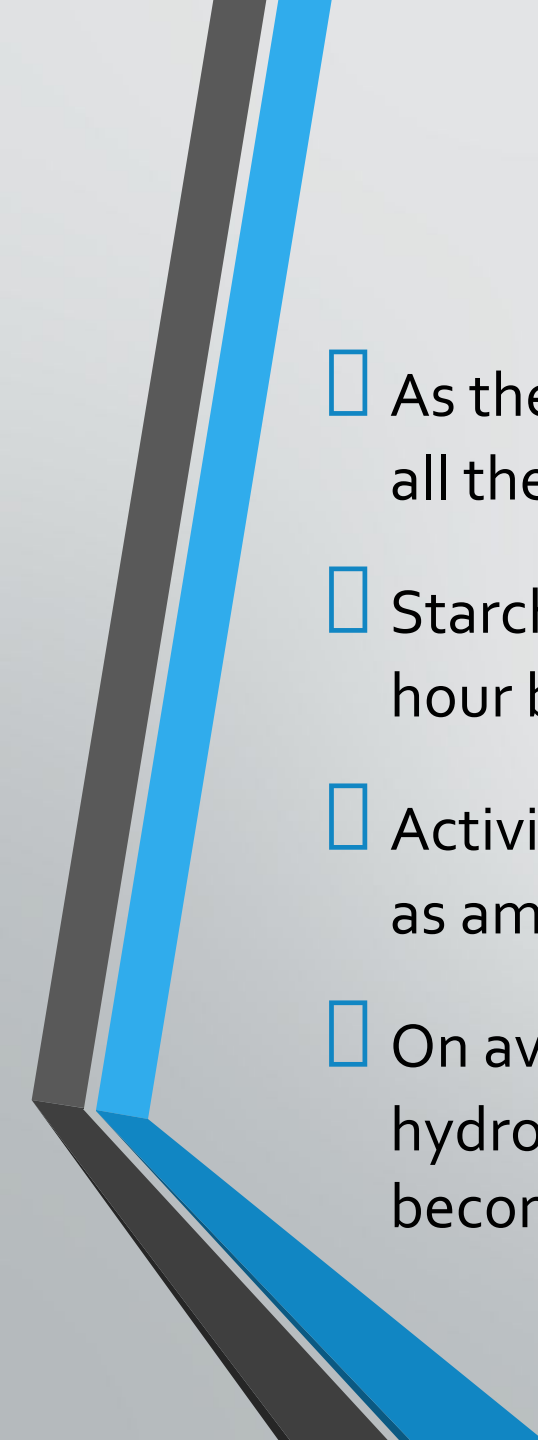




DIGESTION OF CARBOHYDRATES

Digestion of carbohydrates in the mouth & stomach

- Enzymes involved in the digestion of carbohydrates are *amylolytic enzymes*.
- The amylolytic enzyme present in saliva is the *salivary amylase* or *ptyalin*.
- Salivary amylase is secreted mainly by the *parotid glands*.
- This enzyme hydrolyzes starch into the disaccharide *maltose* & other small polymers of glucose that contain three to nine glucose molecules.

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- As the food remains in the mouth only for a short time, not more than 5% of all the starches are hydrolyzed before the food is swallowed.
 - Starch digestion then continues in the ***fundus of stomach*** for as long as 1 hour before the food is mixed with the stomach secretions.
 - Activity of salivary amylase is then blocked by acids of the gastric secretions as amylase becomes inactive when the pH falls below 4.0.
 - On average, as much as 30% to 40% of the starches will have been hydrolyzed mainly to maltose once the food & its accompanying saliva becomes completely mixed with the gastric secretions.

DIGESTION OF CARBOHYDRATES IN THE SMALL INTESTINE

Digestion by pancreatic amylase

- Pancreatic secretions like saliva contains large quantity of alpha-amylase.
- Alpha-amylase present in pancreatic secretions is almost identical in function to salivary amylase but is several times much powerful.
- Within 15 to 30 minutes after the chyme empties from the stomach into the duodenum & mixes with pancreatic juice, almost all the carbohydrates will have been digested.
- Generally, the carbohydrates are almost totally converted into maltose & other small glucose polymers before passing beyond the duodenum & upper jejunum.

Carbohydrate Digestion

1

Mouth: Salivary amylase breaks down food particles and continues to break down through esophagus

2

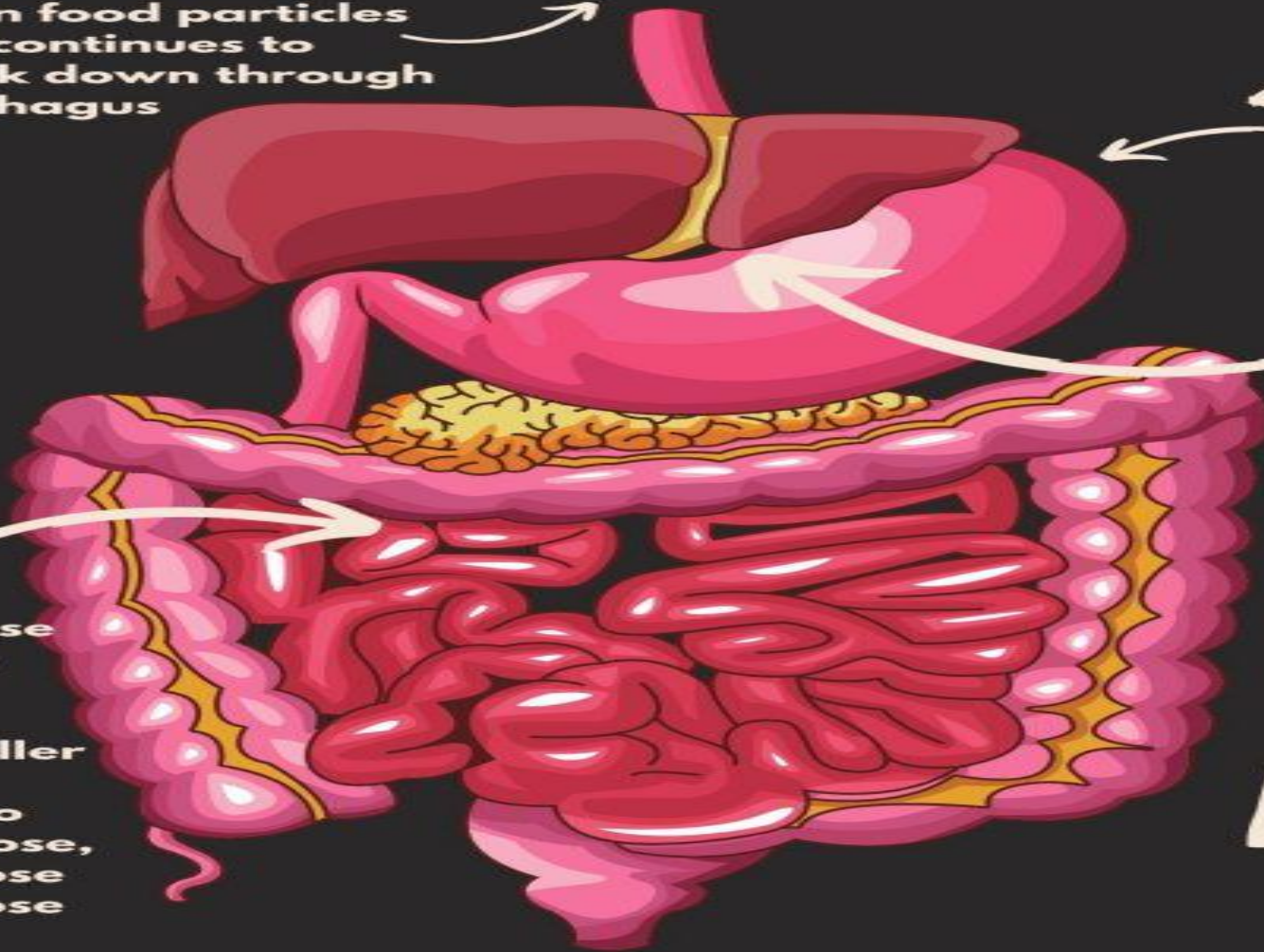
Stomach: Contractions mix the carbohydrates into the more uniform mixture called Chyme

3

Small Intestine: Pancreatic amylase is released which starts the breakdown of dextrins into smaller chains. Disaccharides also break down sucrose, maltose and lactose into smaller glucose molecules for absorption.

Liver: the first organ to receive glucose, fructose, and galactose (the end products of digestion) and glucose is either stored as glycogen or exported back to the blood

4



Hydrolysis of disaccharides & small glucose polymers by intestinal epithelial enzymes

- The enterocytes lining the villi of the small intestine contain enzymes ***lactase, sucrase, maltase & alpha-dextrinase*** which splits the disaccharides into their constituent monosaccharides.
- These enzymes are located in the enterocytes covering the intestinal microvilli brush border.
- The disaccharides are digested as they come in contact with these enterocytes.
- The final products of carbohydrates digestion are monosaccharides that are all ***water-soluble*** & are absorbed immediately into the portal blood.

HOW ARE CARBOHYDRATES DIGESTED?

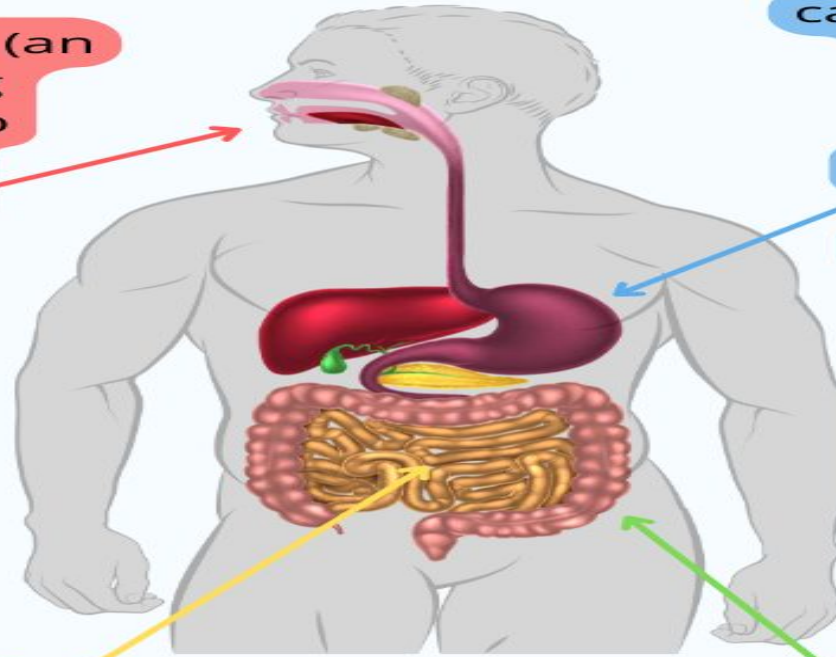
HAVING A BASIC UNDERSTANDING OF DIGESTION CAN ENHANCE YOUR NUTRITIONAL LITERACY

1. MOUTH

Chemical and mechanical digestion begins in the mouth. Salivary amylase (an enzyme) begins breaking down carbohydrates into shorter polysaccharide chains. Chewing breaks down food into more manageable pieces for digestion.

2. STOMACH

Mechanical digestion of carbohydrates continues in the stomach. No further chemical breakdown occurs because salivary amylase is neutralised by our stomach acid (HCl). Food contents are mixed into a substance called chyme.



3. SMALL INTESTINE

The majority of carbohydrate digestion occurs in the small intestine. The enzyme pancreatic amylase is released by the pancreas which breaks down the larger carbohydrate chains. Brush border enzymes help convert these larger chains into simple sugars which are then absorbed into the bloodstream.

4. LARGE INTESTINE

Dietary fibre and other indigestible carbohydrates are metabolised by our gut bacteria. This produces certain byproducts such as short chain fatty acids and gas. The remaining fibre is excreted in the faeces.

ABSORPTION OF CARBOHYDRATES

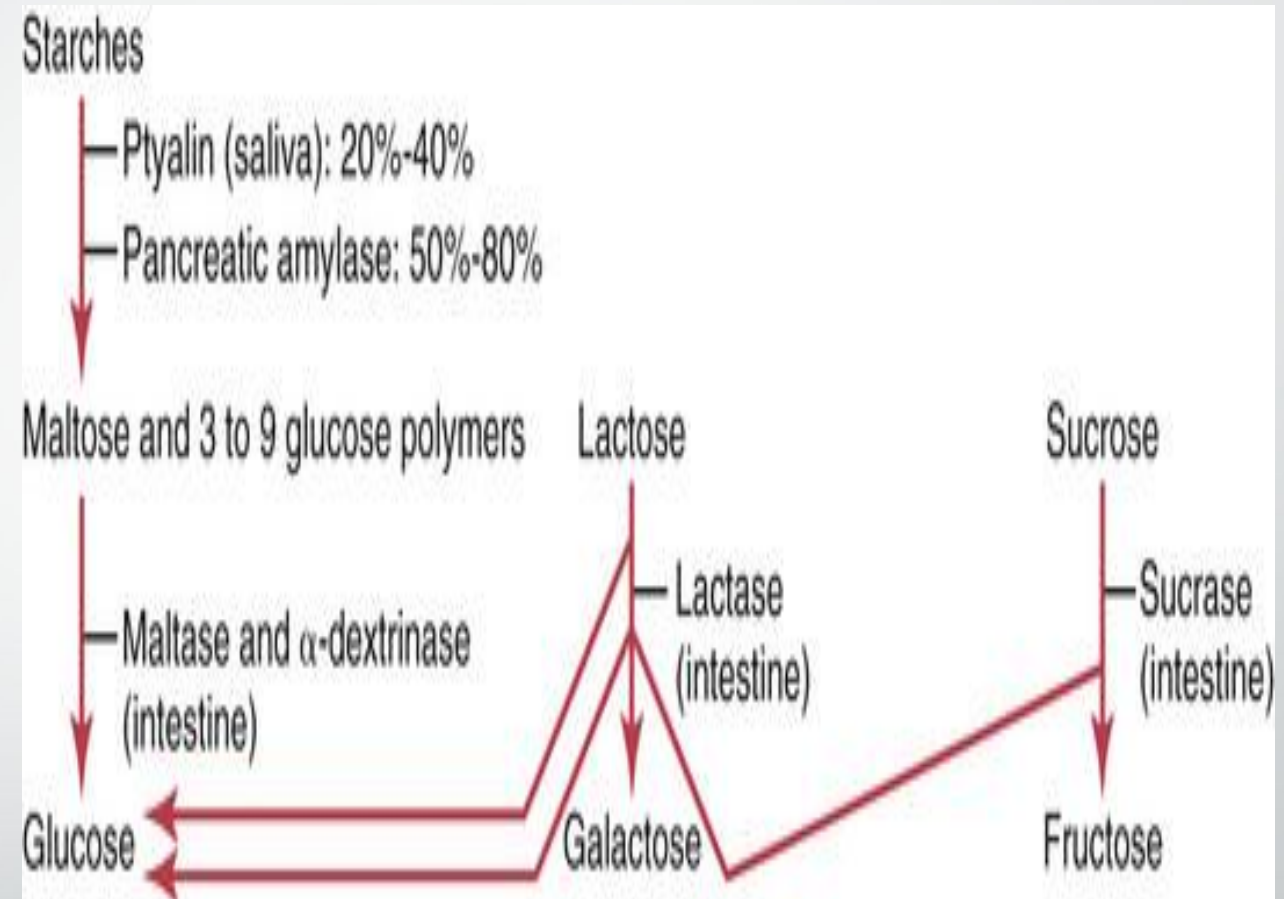
- ❑ Carbohydrates are absorbed from the small intestine mainly as monosaccharides i.e. glucose, galactose & fructose.

Absorption of glucose & galactose

- ❑ Glucose & galactose is transported from the lumen of small intestine into the epithelial cells in the mucus membrane of small intestine by means of ***sodium cotransport***.
- ❑ Energy for this transport is obtained by the binding process of sodium ion & glucose molecule to carrier protein.
- ❑ From the epithelial cells, glucose & galactose is absorbed into the portal vein by ***facilitated diffusion***.
- ❑ However, sodium ion moves laterally into the intercellular space from where it is transported into the blood by active transport utilizing energy liberated by the breakdown of ATP to ADP.

Absorption of fructose

- Fructose is also absorbed from the small intestine into the blood by means of ***facilitated diffusion***.
- However, some molecules of fructose are first converted into glucose & then glucose molecules are absorbed into the blood.



Mouth

Stomach

**Small
intestine**

**Intestinal
lining**

Dietary carbohydrates

Salivary α -amylase

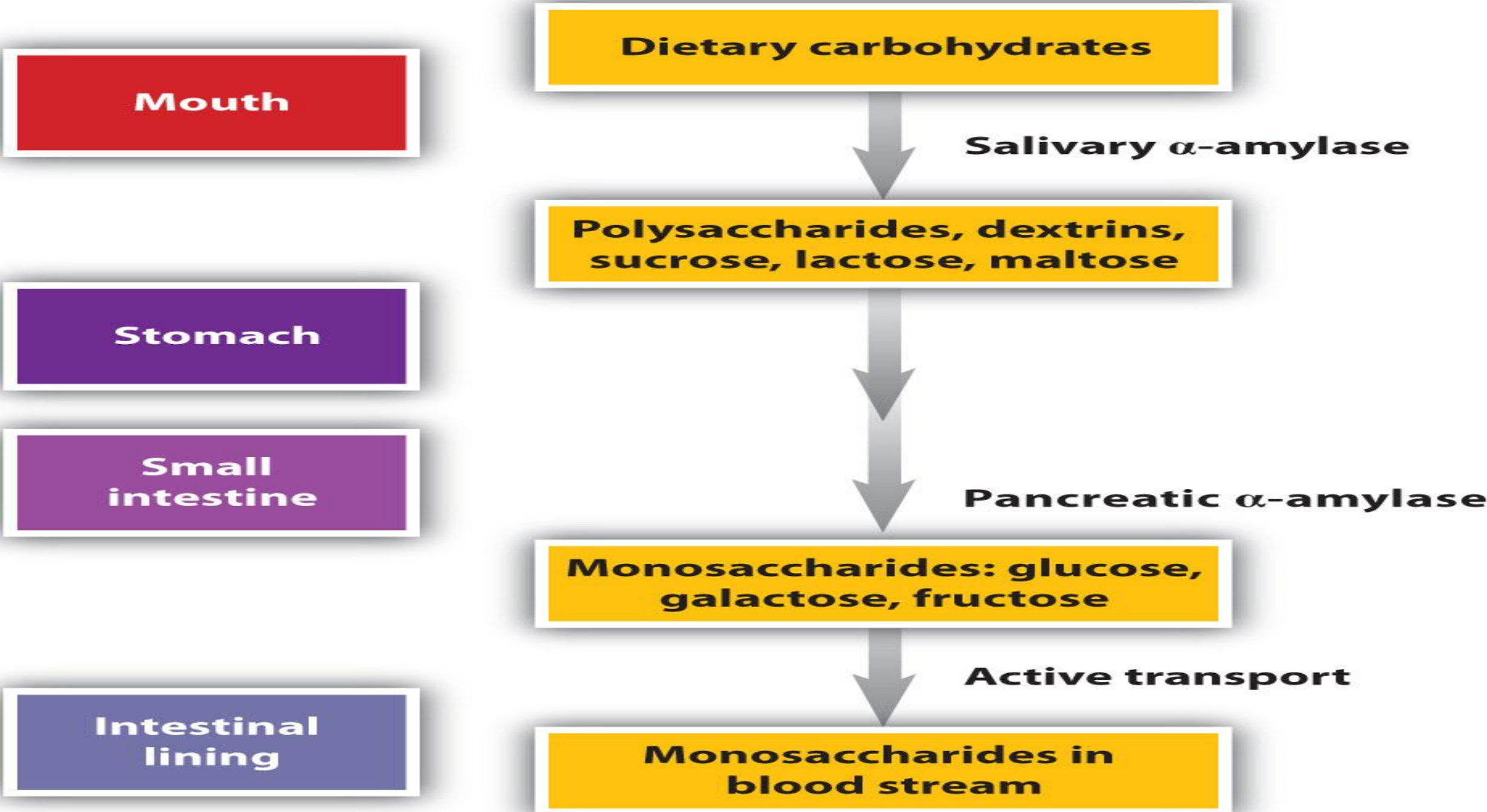
**Polysaccharides, dextrans,
sucrose, lactose, maltose**

Pancreatic α -amylase

**Monosaccharides: glucose,
galactose, fructose**

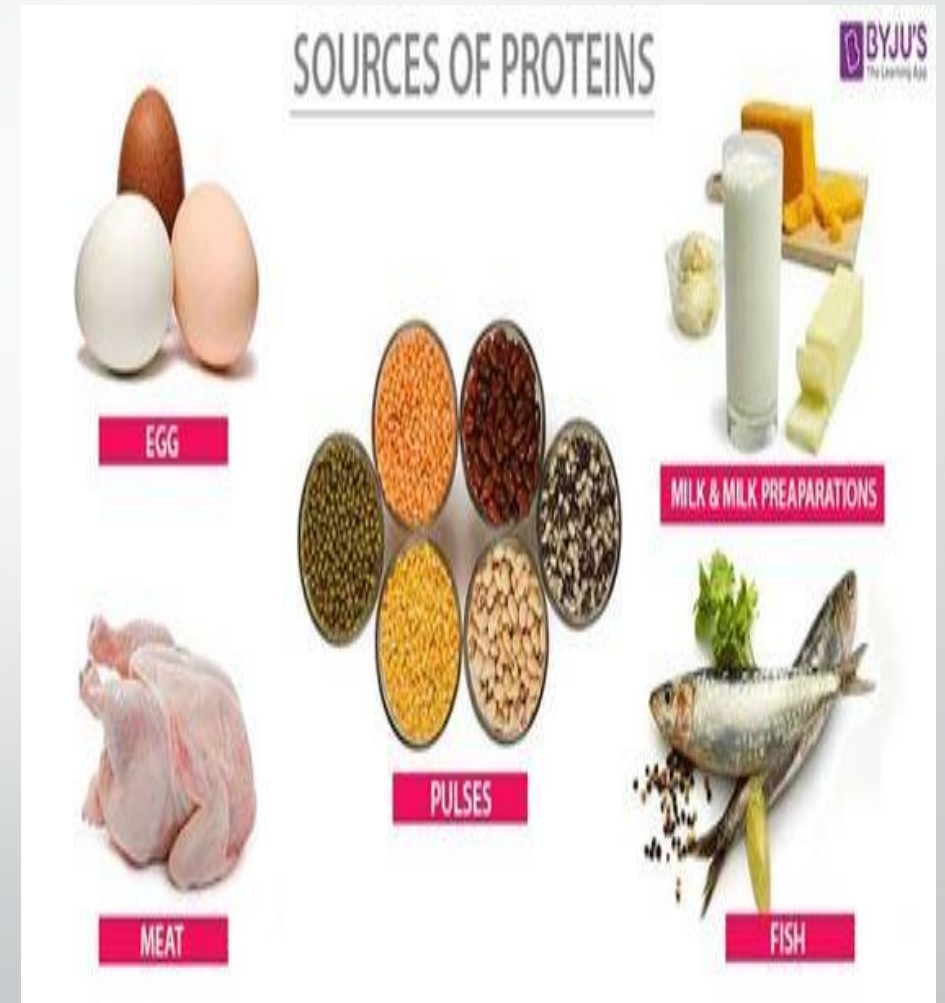
Active transport

**Monosaccharides in
blood stream**



PROTEINS IN DIET

- ❑ Foods that contain high protein content are meat, fish, eggs & milk.
- ❑ Proteins are also present in wheat, soyabeans, oats & different types of pulses.
- ❑ Proteins present in wheat are **glutenin** & **gliadin** which constitute **gluten**.
- ❑ Proteins that are available in milk are **casein**, **lactalbumin**, **albumin** & **myosin**.
- ❑ Eggs also contain proteins as **albumin** & **myosin**.
- ❑ Proteins present in the diet are formed by long chains (polypeptide chains) of amino-acids bound together by peptide linkages.



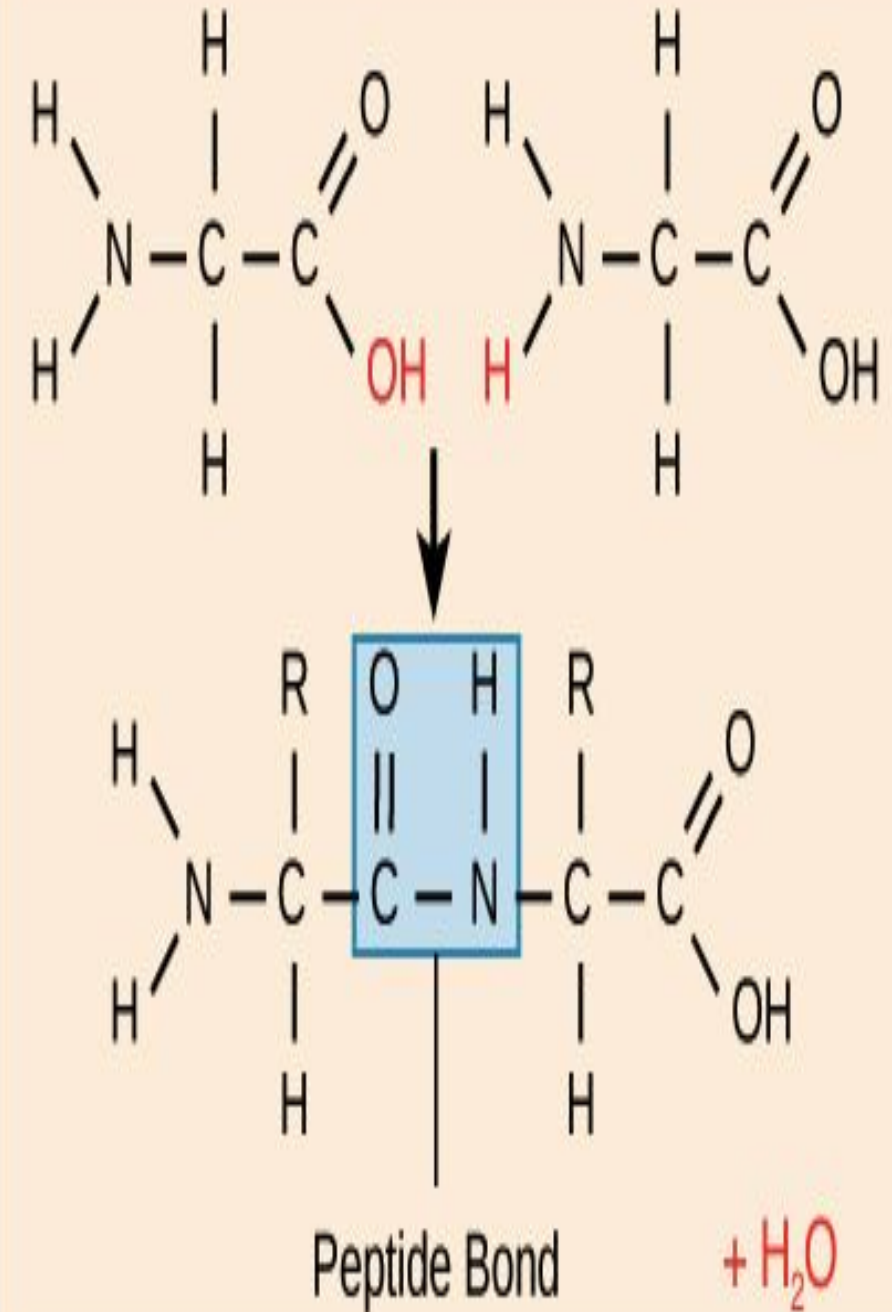
DIGESTION OF PROTEINS

Digestion starts in the stomach

- Enzymes responsible for the digestion of proteins are called *proteolytic enzymes*.
- Digestion of proteins starts in the stomach as saliva does not contain any proteolytic enzyme.
- Pepsin is the only proteolytic enzyme present in the gastric juice.
- Pepsin is most active at pH 2.0 to 3.0 & inactive at pH above about 5.0.
- One important feature of pepsin activity is its ability to digest the protein collagen(an albuminoid type of protein)affected very little by other digestive enzymes.

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- It is necessary that collagen fibers must be digested to penetrate meat & digest other meat proteins because collagen is a major constituent of intercellular connective tissue of meat.
- Pepsin initiates the digestion of proteins contributing only 10% to 20% of total protein digestion.
- It converts proteins to ***proteoses, peptones*** & a ***few polypeptides*** through ***hydrolysis*** at the peptide linkages between the amino-acids.

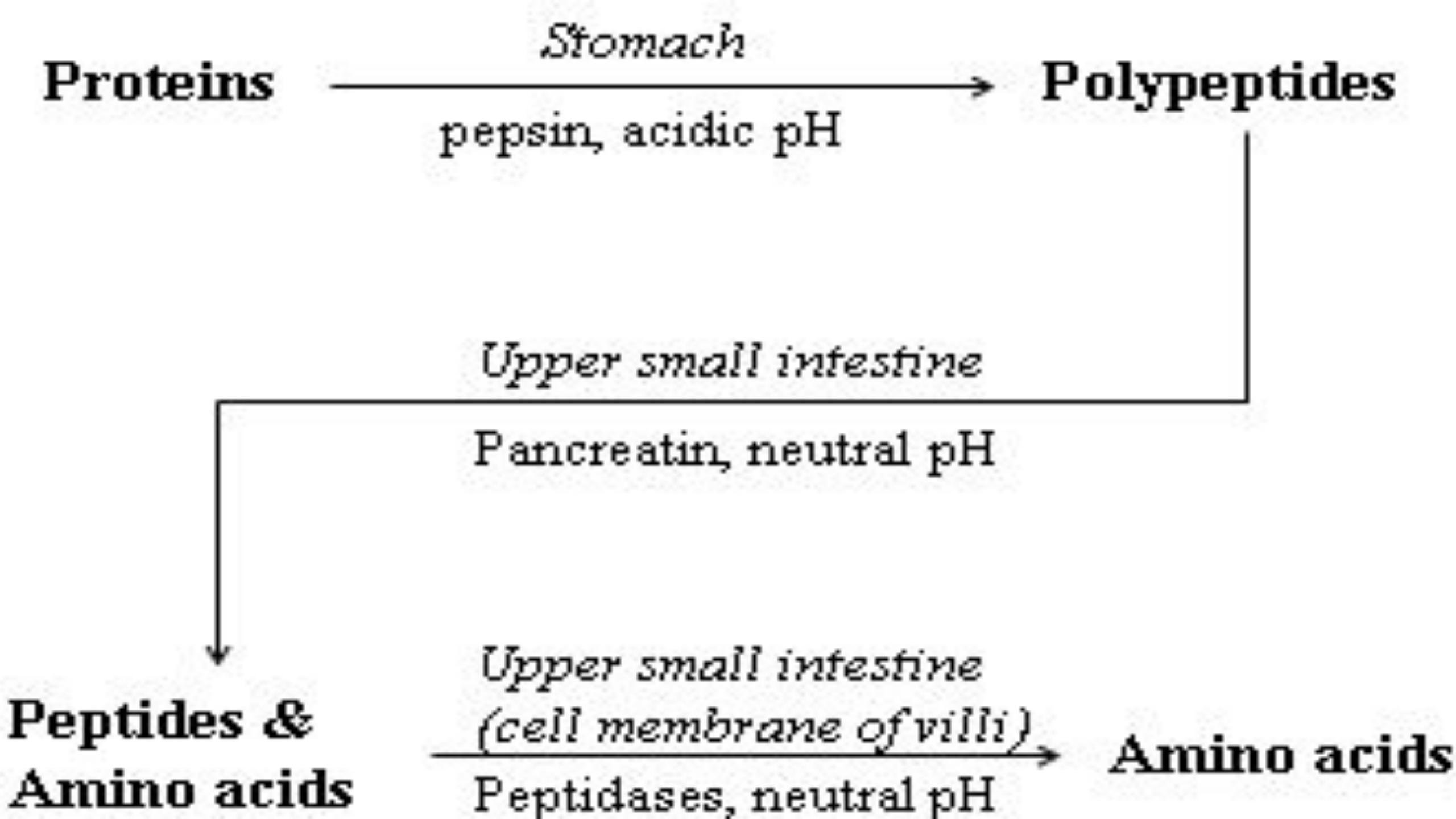


Proteolytic enzymes present in pancreatic juice

- Most protein digestion occurs in the upper small intestine, duodenum & jejunum by proteolytic enzymes of pancreatic secretions.
- Pancreatic juice contains ***trypsin, chymotrypsin, carboxypeptidases & proelastase*** which attack the partial breakdown products of protein food.
- Trypsin & chymotrypsin split proteins into small polypeptides which are then cleaved by carboxypeptidase into individual amino-acids at carboxyl ends of polypeptides.
- Proelastase converted into elastase digest the elastin fibers that partially holds the meat together.
- Only small percentages of proteins are digested to their constituent amino-acids by the pancreatic juices & most remain as ***dipeptides & tripeptides***.

Peptidases in the enterocytes that line the small intestinal villi

- Final digestion of proteins takes place by the proteolytic enzymes present in the ***succus entericus***.
- Two types of peptidase enzymes are especially important i.e. ***aminopolypeptidase*** & several ***dipeptidases*** that split the remaining larger polypeptides into tripeptides & dipeptides & a few into their constituent amino-acids.
- The amino-acids, dipeptides & tripeptides are then easily transported through the microvillar membrane to the interior of enterocytes.
- Finally, inside the enterocyte are multiple other peptidases specific for the remaining types of linkages between amino-acids.

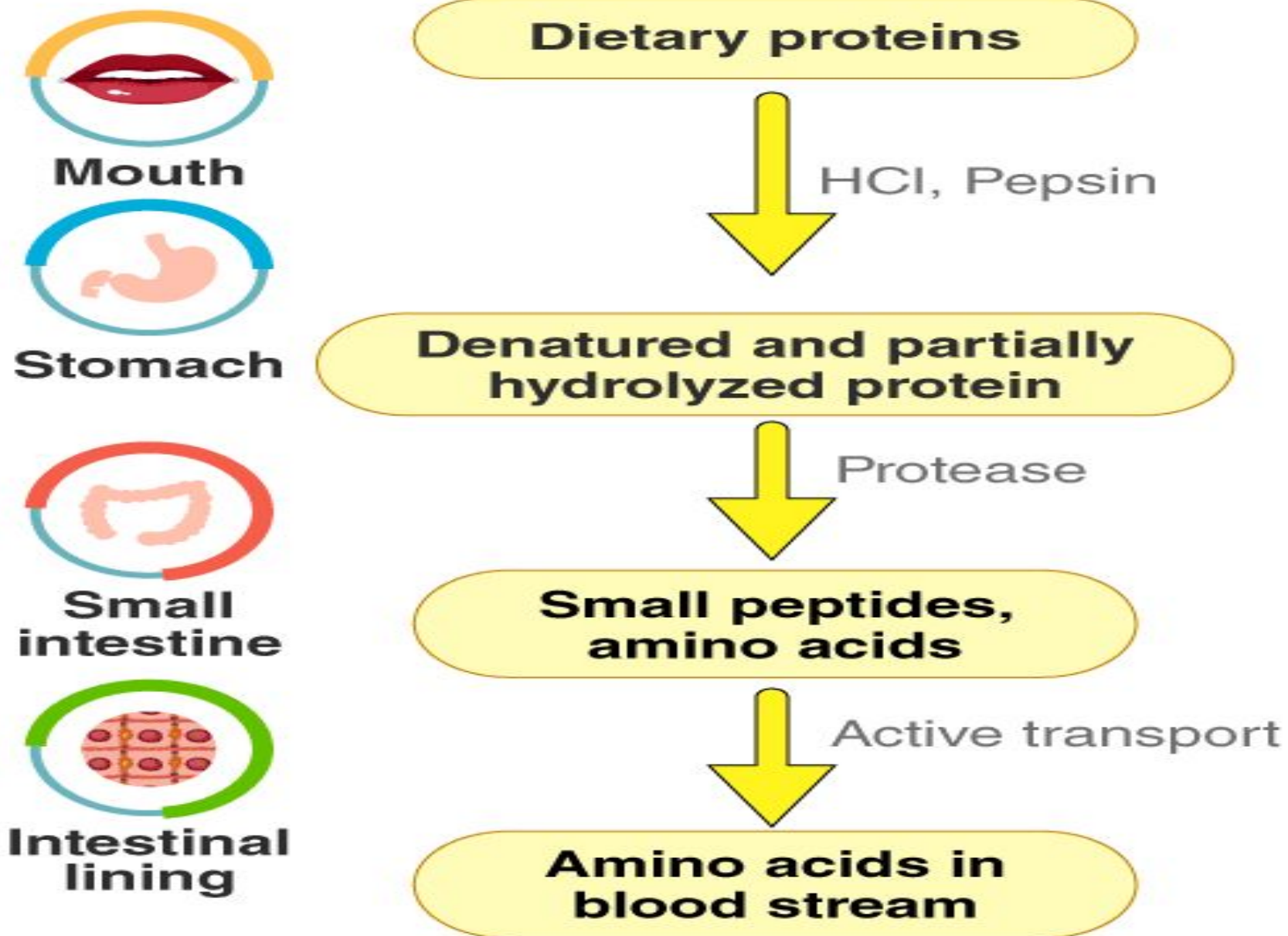


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- More than 99% of final protein digestive products that are absorbed are individual amino-acids with only rare absorption of peptides & very rare absorption of whole protein molecules.
- These few absorbed molecules of whole protein can even sometimes cause serious allergic & immunologic disturbances.

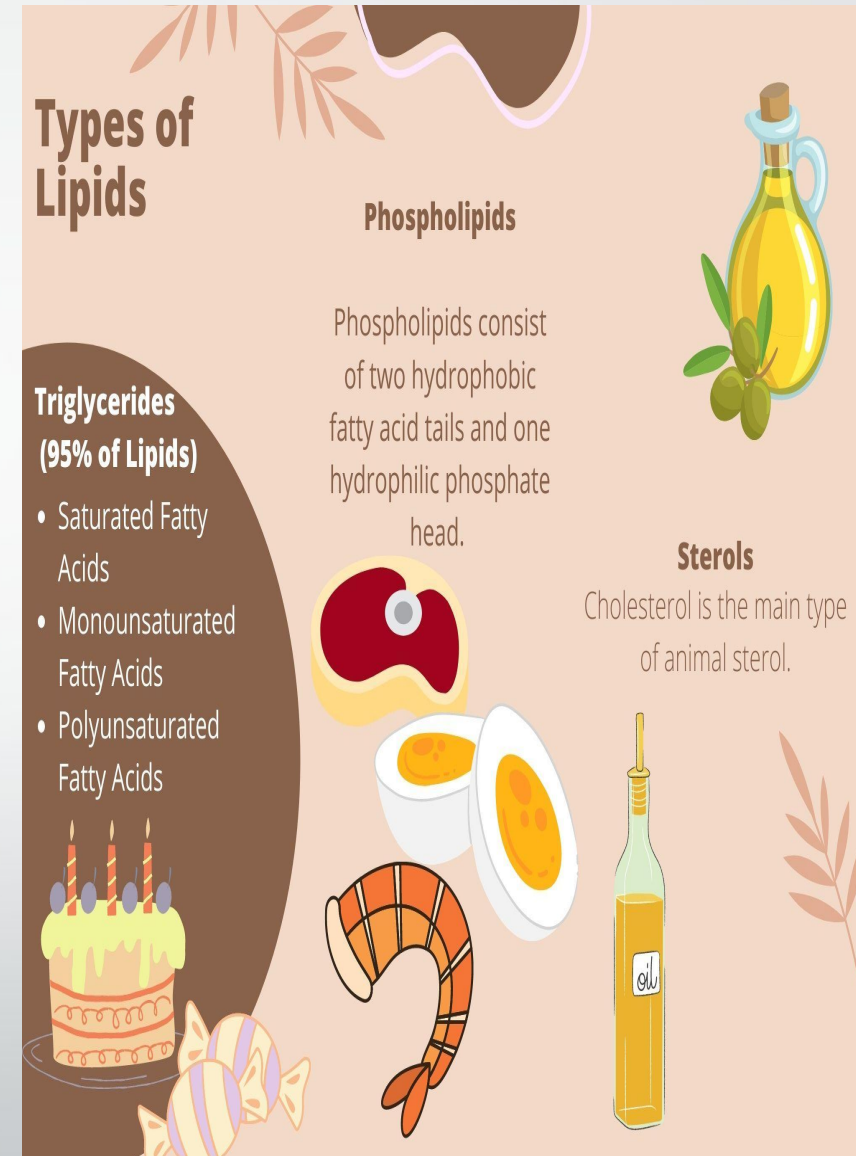
ABSORPTION OF PROTEINS

- Proteins are absorbed in the form of amino-acids from small intestine.
- The ***levo amino-acids*** are actively absorbed by means of ***sodium cotransport*** & the ***dextro amino-acids*** are absorbed by means of ***facilitated diffusion***.



FATS IN THE DIET

- ❑ Fats are mostly consumed in the form of **neutral fats** also known as **triglycerides**.
- ❑ Triglycerides are made up of **glycerol** & **free fatty acids**.
- ❑ Triglycerides form the major fat constituent in foods of animal origin & much less in foods of plant origin.
- ❑ Apart from triglycerides, usual daily diet also contains small quantities of **phospholipids**, **cholesterol** & **cholesterol esters**.
- ❑ Cholesterol is a sterol compound & contains no fatty acid but exhibit some physical & chemical characteristics of fats.
- ❑ Also, it is derived from fats & is metabolized similarly to fats therefore cholesterol is considered to be a fat.



SATURATED & UNSATURATED FATS

- Fats which contain triglycerides formed from only saturated fatty acids are called ***saturated fats***.
- Saturated fats have maximum amount of hydrogen ions without any double bonds between carbon atoms.
- Fats containing unsaturated fatty acids are known as ***unsaturated fats***.
- Unsaturated fatty acids are formed by the ***dehydrogenation*** of saturated fatty acid.
- Unsaturated fats are of three types:
 - I. ***Monounsaturated fats***(one double bond between carbon atoms).
 - II. ***Polyunsaturated fats***(more than one double bond between carbon atoms).
 - III. ***Trans fats***(across or opposite side double bonds between carbon atoms).

THE DIETARY FATS

@cheatdaydesign

POLYUNSATURATED FAT

An essential fat (we must get from food because our bodies cannot produce) & lowers LDL (bad cholesterol)

Found in: Most cooking oils, pumpkin seeds, pine nuts, sesame seeds, fatty fish

Also known as: Omega-3 and Omega-6 fatty acids



MONOUNSATURATED FAT

Considered a healthy fat: Lowers LDL (bad cholesterol) & maintains HDL (good cholesterol)

Found in: Olive oil, avocado & avocado oil, most nuts & nut butters



SATURATED FAT

Increases total cholesterol & LDL (bad cholesterol). Best to consume in moderation.

Found in: Red meat, whole milk, cheese, coconut, butter, processed meat, many baked goods, deep fried foods



TRANS FAT

A by-product of processing healthier fats to give them a longer shelf life. Raises your LDL (bad cholesterol) and lowers your HDL (good cholesterol). Try to limit.

Also known as: Partially hydrogenated oil

**Previously found in margarine & many processed foods, but has been banned by the FDA in the U.S. as of 2018*



DIGESTION OF FATS

Starts in the stomach but occurs mainly in the small intestine

- A small amount of triglycerides is digested in the stomach by *lingual lipase* secreted by the lingual glands in the mouth & swallowed with saliva.
- This amount of digestion is less than 10% & essentially all fat digestion occurs in the small intestine.

Emulsification by bile acids & lecithin

- Emulsification is a process of physical breaking of fat globules into small size so that the water-soluble digestive enzymes can act on the globule surface.

CONT..

- Emulsification begins by agitation in the stomach to mix the fat with the products of stomach digestion.
- Most of the emulsification then occurs in the duodenum under the influence of bile(the liver secretion that does not contain any digestive enzyme).
- However, bile contains a large quantity of bile salts & the phospholipid lecithin which is particularly important for emulsification of the fats.
- The polar parts(points where ionization occurs in water)of the bile salts & lecithin molecules are highly soluble in water whereas most of the remaining portions of their molecules are highly soluble in fat.
- Therefore, the fat-soluble portions of these liver secretions dissolve in the surface layer of fat globules with the projecting polar portions.

CONT..

- These polar projections are soluble in the surrounding watery fluids which greatly decreases the interfacial tension of the fat & makes it soluble as well.
- When the interfacial tension of a globule of non miscible fluid is low, agitation can break this non miscible fluid into many tiny particles much more easily.
- So the major function of bile salts & lecithin in the bile is to make fat globules readily fragmentable by agitation with water in the small intestine causing a total of 1000-fold increase in the surface area of fats by emulsification.
- The lipase enzymes are water-soluble & can attack the globules only on their surface.

TRIGLYCERIDES DIGESTED BY PANCREATIC LIPASE

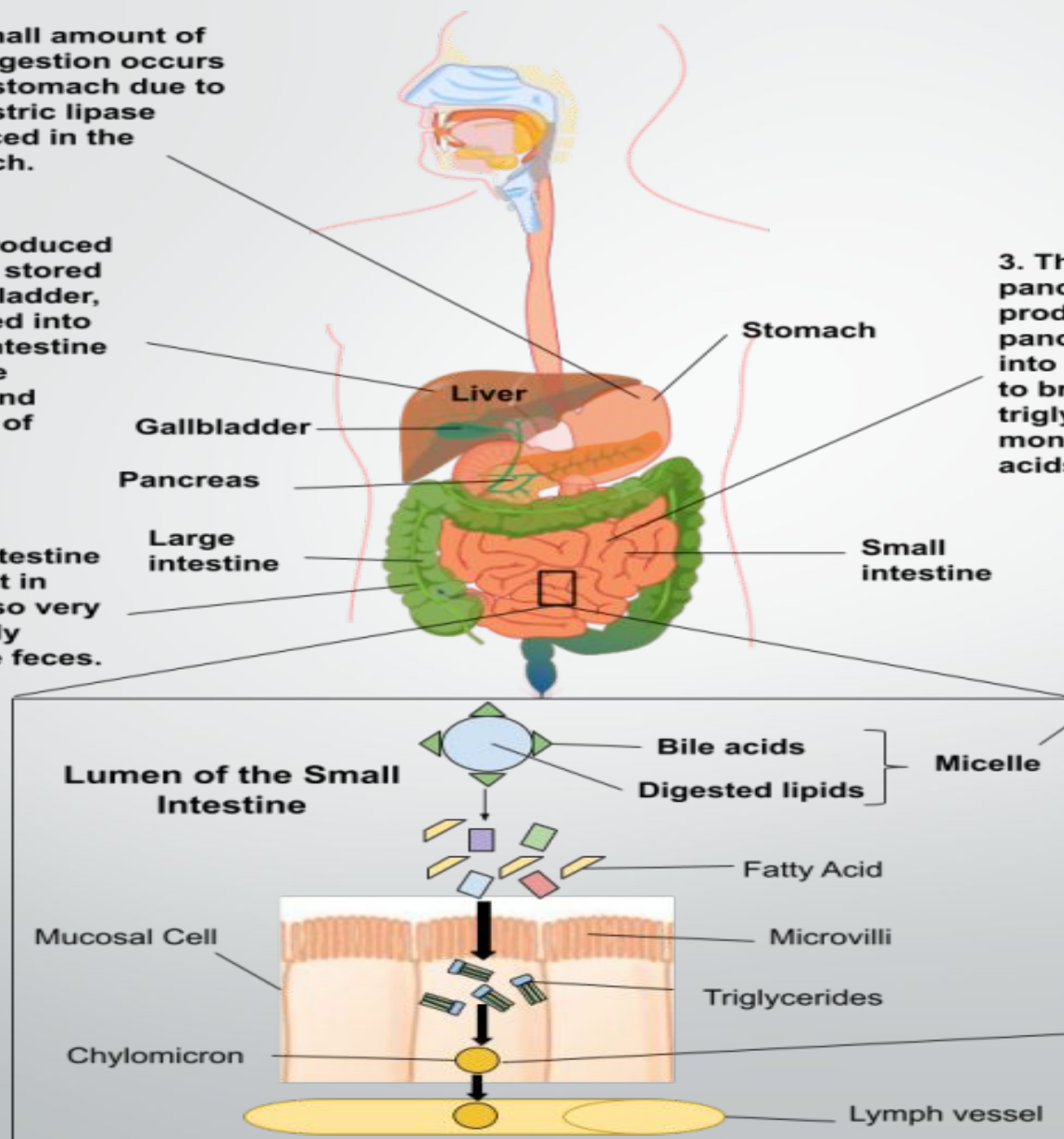
- The most important enzyme for the digestion of triglycerides is ***pancreatic lipase*** present in enormous quantities in pancreatic juice enough to digest all triglycerides & split them into free fatty acids & 2-monoglycerides.
- The enterocytes of the small intestine contain additional lipase called ***enteric lipase*** but it is usually not needed.
- The final products of fat digestion are ***free fatty acids & monoglycerides***.

1. A small amount of lipid digestion occurs in the stomach due to the gastric lipase produced in the stomach.

2. Bile is produced in the liver, stored in the gallbladder, and released into the small intestine to aid in the digestion and absorption of lipids.

3. The enzyme pancreatic lipase is produced by the pancreas and released into the small intestine to break down triglycerides into monoglycerides, fatty acids, and glycerol.

6. The small intestine is very efficient in absorbing fat so very little is normally excreted in the feces.



4. In the small intestine, the products from fat digestion and bile acids form a micelle, which moves toward the microvilli to allow the lipids to diffuse into the mucosal cells.

5. Inside the mucosal cells, the fatty acids and monoglycerides are reassembled into triglycerides and incorporated into lipid transport particles called chylomicrons, which enter the lymph vessel.

DIGESTION OF CHOLESTEROL ESTERS & PHOSPHOLIPIDS

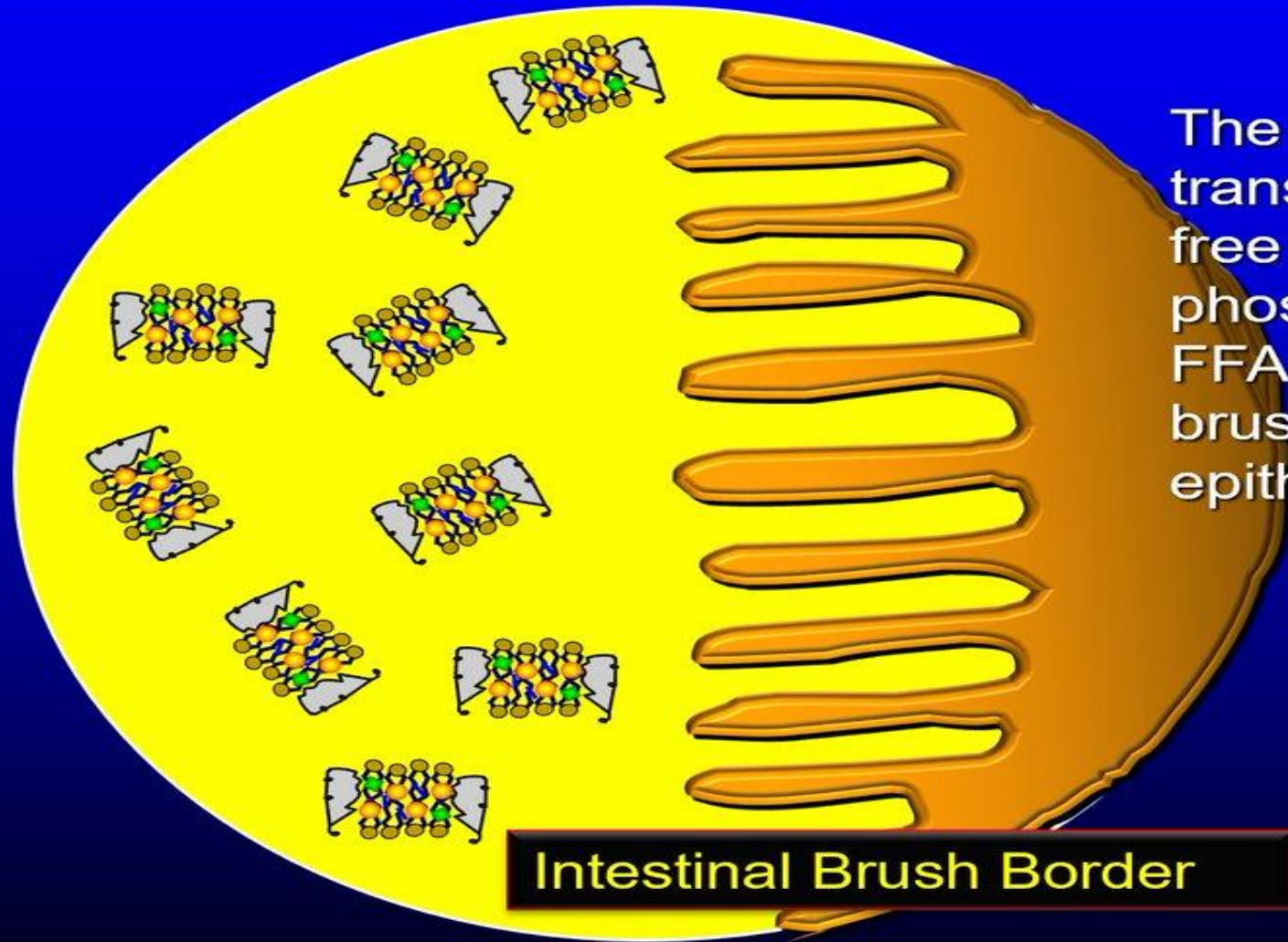
- Most cholesterol in the diet is in the form of cholesterol esters which are the combinations of free cholesterol & one molecule of fatty acid.
- The enzyme ***cholesterol ester hydrolase*** present in pancreatic secretion hydrolyze the cholesterol esters of the diet.
- Phospholipids also contain fatty acids within their molecules & is hydrolyzed by ***phospholipase A₂*** present in the pancreatic secretion.

BILE SALTS MICELLES ACTS AS A "FERRY"

- ❑ Micelles are small, spherical, cylindrical globules 3 to 6 nanometers in diameter & is composed of 20 to 40 molecules of bile salts.
- ❑ Each bile salt molecule is composed of a sterol nucleus(fat-soluble) & a polar group(water-soluble).
- ❑ The sterol nucleus encompasses the fat digestate forming a fat globule in the middle with polar groups of bile salts projecting outward to cover the surface of micelle.
- ❑ These bile salts micelles act as a transport medium to carry the monoglycerides & free fatty acids which would otherwise be relatively insoluble to the brush borders of the intestinal epithelial cells.
- ❑ From here, the monoglycerides & free fatty acids are absorbed & the bile salts are released back into the chyme to be used again for the same process.
- ❑ The bile salts micelles play the same role in transporting free cholesterol & phospholipids molecules digestates also.

Luminal Phase of Lipid Absorption

Micelle Delivery at Brush Border



The **bile salt micelles** act as a transport medium to carry or “ferry” free cholesterol, phytosterols, phospholipid, monoglycerides and FFA including omega-3 FFA to the brush borders of the intestinal epithelium

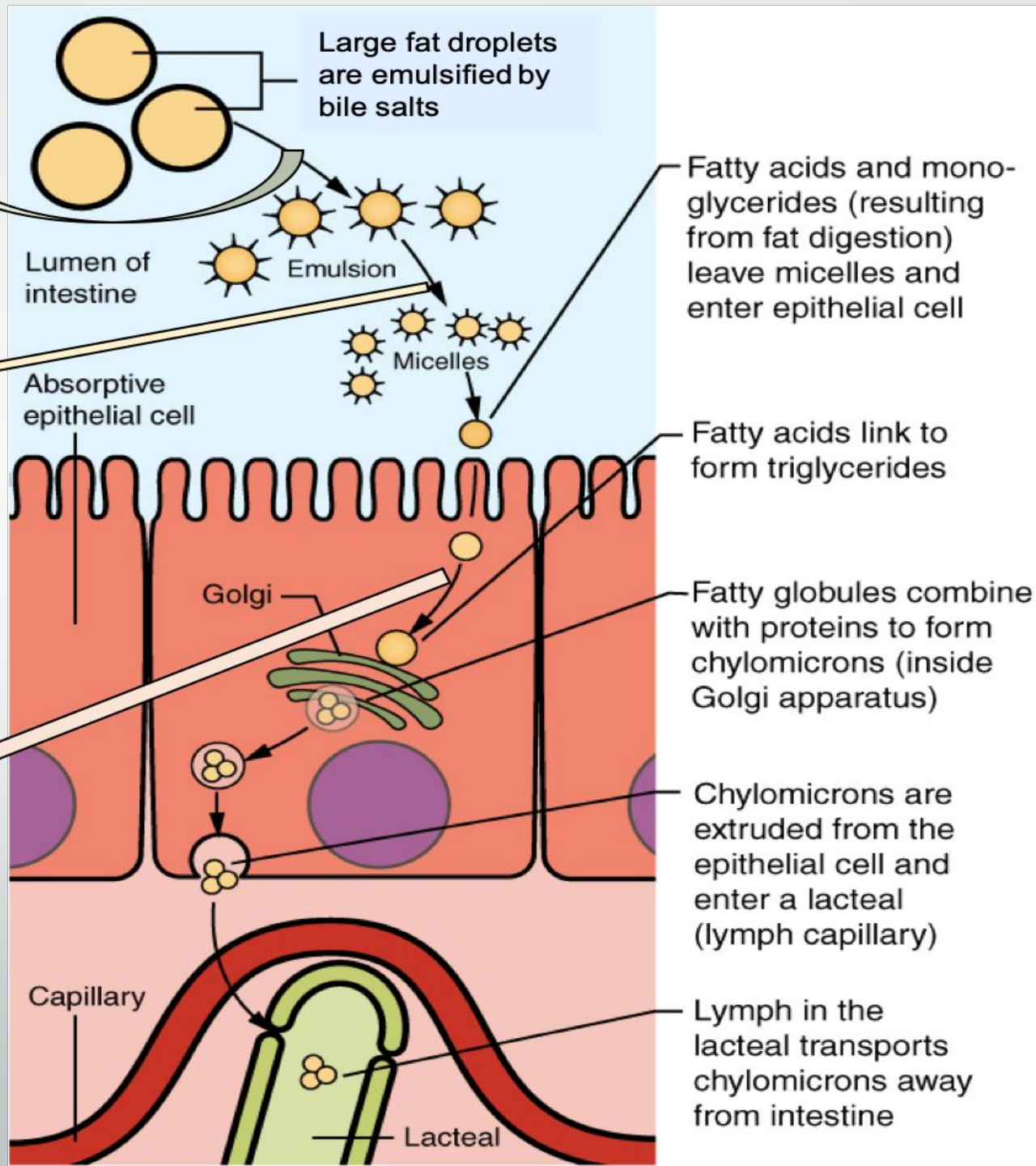
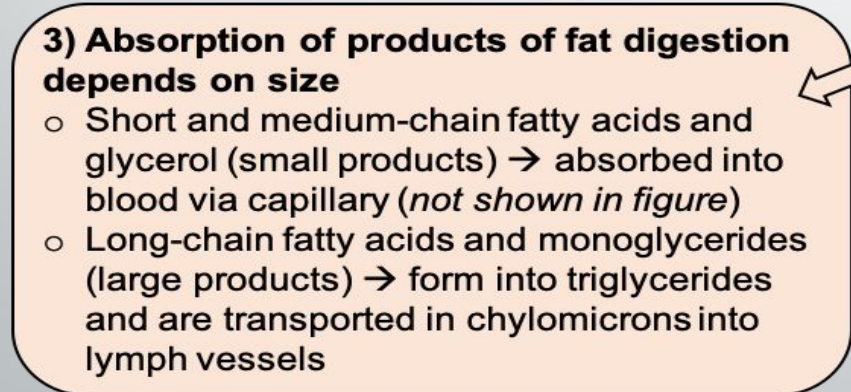
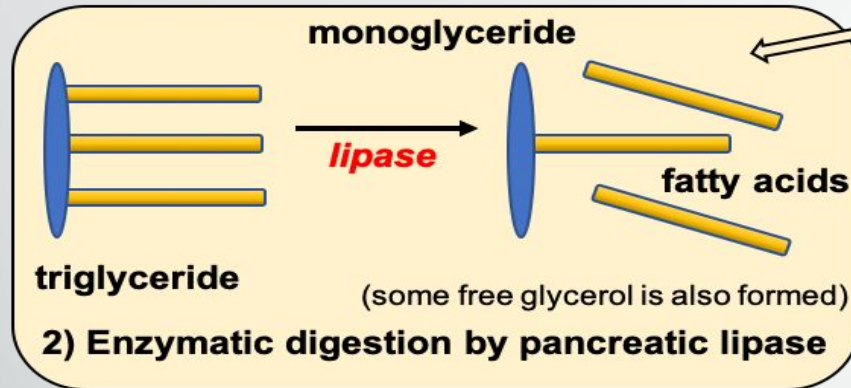
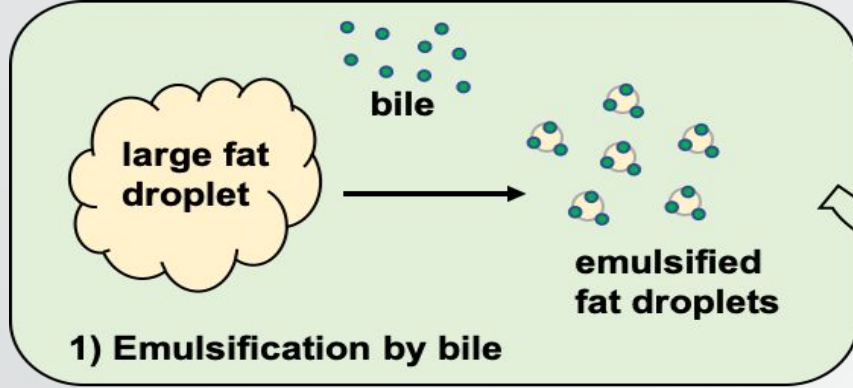
Intestinal Brush Border

ABSORPTION OF FATS

- The micelles performing the “ferrying” function are highly important for fat absorption.
- In the presence of abundant bile micelles, about 97% of the fat is absorbed whereas in the absence of bile micelles only 40% to 50% can be absorbed.
- The monoglycerides & free fatty acids carried to the surface of microvilli of intestinal brush border by the bile salts micelles penetrate into the recesses among the moving microvilli.
- From here, both the monoglycerides & fatty acids diffuse immediately out of the micelles into the interior of epithelial cells as lipids are soluble in the epithelial cell membrane.

In the form of '*chylomicrons*' through the lymph

- ❑ After entering the epithelial cell, the fatty acids & monoglycerides are taken up by the cell's smooth endoplasmic reticulum.
- ❑ Here, they are mainly used to form new triglycerides that are subsequently released in the form of chylomicrons.
- ❑ Chylomicrons are the triglycerides & cholesterol esters coated with a layer of protein, cholesterol & phospholipids.
- ❑ These chylomicrons cannot pass through the membrane of blood capillaries because of their larger size.
- ❑ So these lipid particles(chylomicrons) enter the lymph vessels & then transferred into the blood from lymph.



Direct absorption of short & medium chain fatty acids into portal blood

- Small quantities of short & medium chain fatty acids(e.g. from butter fat) are absorbed directly into portal blood.
- As short chain fatty acids are more water soluble so mostly not converted into triglycerides by the endoplasmic reticulum.
- Therefore the short chain fatty acids diffuse directly from the intestinal epithelium into the capillary circulation of intestinal villi.

POST ASSESSMENT

Q1. Which enzyme initiates protein digestion in the stomach?

- a) Trypsin
- b) Pepsin
- c) Chymotrypsin
- d) Amylase
- e) Lipase

Q2. Bile salts are essential for:

- a) Protein digestion
- b) Carbohydrate digestion
- c) Vitamin B-12 absorption
- d) Emulsification of fats
- e) Water absorption

Q3. A patient with pancreatic insufficiency is most likely to have difficulty absorbing:

- a) Water
- b) Sodium
- c) Lipids
- d) Fructose
- e) Vitamin-E

Q4. Which of the following is the final product of carbohydrates digestion that is absorbed by the enterocytes?

- a) Starch
- b) Glycogen
- c) Disaccharides
- d) Monosaccharides
- e) Polysaccharides